

Licci numerical semigroup ideals

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VSGS 2026 - 18 June 2026

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Let $R = \mathbb{C}[x_1, \dots, x_n]$. Let I be an ideal of R and denote the minimal free resolution of the R -module R/I as

$$0 \rightarrow R^{\beta_n} \longrightarrow R^{\beta_{n-1}} \longrightarrow \cdots \longrightarrow R^{\beta_1} \longrightarrow R^{\beta_0} \longrightarrow \frac{R}{I} \rightarrow 0.$$

The positive integers $\beta_0, \beta_1, \dots, \beta_n$ are the **total Betti numbers** of I and we always have $\beta_0 = 1$. The $(n+1)$ -tuple $(1, \beta_1, \dots, \beta_n)$ is called **format** of the resolution.

An ideal I of R is said to be **perfect** if $n = \text{ht}(I)$, or equivalently, if R/I is a Cohen-Macaulay ring.

The problem of characterizing perfect ideals

A characterization of perfect ideals is still unknown, but the problem is solved if the height is 2 or if the ideal is Gorenstein of height 3:

- the structure of perfect ideals of height 2 is described by the Hilbert-Burch Theorem;
- the structure of perfect ideals I of height 3 such that R/I is a Gorenstein ring was described by Buchsbaum and Eisenbud.

Regular sequences and linkage

A sequence $\{f_1, \dots, f_n\}$ in R is **regular** if

- f_1 is a regular element in R ;
- the class of f_i is a regular element in $R/(f_1, \dots, f_{i-1})$ for all $i = 2, \dots, n$.

The variables $\{x_1, \dots, x_n\}$ form a regular sequence in R .

An ideal is called **complete intersection** if its generators form a regular sequence in R .

Let I be a perfect ideal of R with $\text{ht}(I) = c$. An ideal J is **linked** to I if

$$J = (\alpha) : I = \{x \in R : xI \subseteq (\alpha)\}$$

where α is a regular sequence of length c . In this case J is perfect of height equal to c .

Linkage classes and licci ideals

Two ideals I and J are in the same **linkage class** if it is possible to obtain one from the other by a finite sequence of linked ideals.

Being in the same linkage class is an equivalence relation.

An ideal I of R is **licci** if it is in the same linkage class of a complete intersection ideal.

If I is a perfect ideal and $\text{ht}(I) \leq 2$, then I is licci, but this is false if $\text{ht}(I) \geq 3$. For instance, the ideal

$$(x, y, z)^2 \subseteq \mathbb{C}[x, y, z]$$

is perfect but not licci (this follows by a criterion by Huneke and Ulrich).

To classify licci ideals is another classical problem in commutative algebra.

Toric ideal defined by a numerical semigroups

Let S be a numerical semigroup minimally generated by $n_0 < n_1 < \cdots < n_{e-1}$. Let $R = \mathbb{C}[X_0, \dots, X_{e-1}]$ graded by

$$\deg(X_i) = n_i \quad \forall i = 0, \dots, e-1.$$

Consider the map

$$\begin{aligned} \varphi : R &\longrightarrow \mathbb{C}[t] \\ X_i &\longrightarrow t^{n_i} \quad \forall i = 0, \dots, e-1 \end{aligned}$$

then, the **defining toric ideal** of S is

$$I_S = \ker(\varphi)$$

and the **semigroup ring** of S is

$$\mathbb{C}[S] = \text{im}(\varphi) \cong \mathbb{C}[X_0, \dots, X_{e-1}] / I_S.$$

Let S be a numerical semigroup minimally generated by e elements. Then the ideal I_S is perfect, since R/I_S is a one-dimensional domain which is not a field (and therefore it is Cohen-Macaulay), and $\text{ht}(I_S) = e - 1$.

Question: When is I_S licci?

We have that I_S is licci if and only if $(X_0) + I_S$ is licci by a result of Huneke and Ulrich.

Artinian algebra associated with a numerical semigroup

Denote by $\mathcal{S}_m$ the set of numerical semigroups with multiplicity m .
Let $S \in \mathcal{S}_m$. Consider the Artinian $\mathbb{C}$ -algebra

$$A(S) := \frac{\mathbb{C}[S]}{t^m \mathbb{C}[S]}.$$

Definition (Kunz)

Two numerical semigroups $S, T \in \mathcal{S}_m$ are said to be **similar** if

$$A(S) \cong A(T)$$

and the isomorphism maps homogeneous elements to homogeneous elements.

Remark

We have that

$$A(S) \cong \frac{R}{(X_0) + I_S} \cong \frac{R/(X_0)}{((X_0) + I_S)/(X_0)}.$$

Similar numerical semigroups

This implies that $(X_0) + I_S$ is licci in R if and only if $\frac{(X_0) + I_S}{(X_0)}$ is licci in $R/(X_0)$. From this one can deduce the following Corollary.

Corollary

If $S, T \in \mathcal{S}_m$ are similar numerical semigroup, then I_S is licci if and only if I_T is licci.

Theorem (Kunz)

Let $S, T \in \mathcal{S}_m$. If S and T are similar, then

- S and T have the same embedding dimension;*
- $A(S)$ and $A(T)$ have the same Betti numbers.*

In particular, since $A(S)$ is obtained from $\mathbb{C}[S]$ as the quotient by a regular element, if S and T are similar, the minimal free resolutions of I_S and I_T have the same Betti numbers.

Kunz cone

Let $S \in \mathcal{S}_m$ and $\text{Ap}(S, m) = \{0, w_1, \dots, w_{m-1}\}$ be the Apéry set, where $w_i \equiv i \pmod{m}$ for every $i = 1, \dots, m-1$.

For every $i \in \{1, \dots, m-1\}$ write

$$w_i = z_i \cdot m + i \quad \forall i = 1, \dots, m-1.$$

The positive integers $z_1, \dots, z_{m-1}$ are called **Kunz coordinates** of S .

The **Kunz cone** $P_m \subseteq \mathbb{R}_{\geq 0}^{m-1}$ is the polyhedral cone with facet inequalities

$$\begin{cases} z_i + z_j \geq z_{i+j} & \text{if } i+j < m, \\ z_i + z_j \geq z_{i+j} - 1 & \text{if } i+j > m. \end{cases}$$

for all $i, j \in \{1, \dots, m-1\}$ such that $1 \leq i \leq j \leq m-1$ such that $i+j \neq m$.

The integral points in P_m with positive coordinates parametrize numerical semigroup with multiplicity m , in other words, a point lies in P_m if and only if it is the vector of Kunz coordinates of a numerical semigroup in $\mathcal{S}_m$.

We identify any numerical semigroup $S \in \mathcal{S}_m$ with the vector of its Kunz coordinates.

Proposition (Kunz)

If $S, T \in \mathcal{S}_m$ lie in the interior of the same faces of P_m , then S and T are similar.

Therefore, being similar is a coarser relation than being in the interior of the same face of P_m .

Hasse diagram of the Apéry set

Let $S \in \mathcal{S}_m$. Consider the usual order relation on S defined as

$$u \leq_S v \quad \stackrel{\text{def}}{\iff} \quad v - u \in S$$

for all $u, v \in S$.

Let $u, v \in S$. We say that v covers u if $v - u$ is a minimal generator of S , and in this case we write $u \sqsubset_S v$.

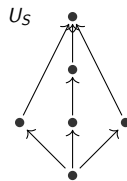
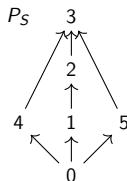
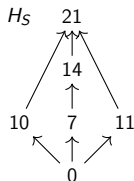
The Hasse diagram of $\text{Ap}(S, m)$ is the directed graph whose set of vertices is $\text{Ap}(S, m)$ and such that (u, v) is an edge if and only if $u \sqsubset_S v$.

Denote

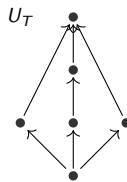
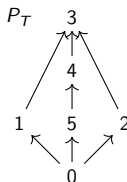
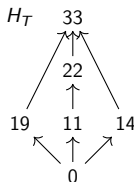
- by H_S the Hasse diagram of $\text{Ap}(S, m)$;
- by P_S the graph obtained by H_S by replacing w_i with i for every $i = 0, \dots, m-1$;
- by U_S the underlying directed graph of H_S (and P_S).

Example (part 1)

Let $S = \langle 6, 7, 10, 11 \rangle$. Then $\text{Ap}(S, 6) = \{0, 7, 14, 21, 10, 11\}$ and



Let $T = \langle 6, 11, 14, 19 \rangle$. Then $\text{Ap}(T, 6) = \{0, 19, 14, 33, 22, 11\}$ and



What information do P_S and U_S encode?

Theorem (Bruns, García-Sánchez, O'Neil, Wilburn, 2019)

Two numerical semigroups $S, T \in \mathcal{S}_m$ lie on the same faces of P_m if and only if $P_S = P_T$.

In particular, P_S encodes the relations that hold in $\text{Ap}(S, m)$.

Proposition

Two numerical semigroups $S, T \in \mathcal{S}_m$ are similar if and only if $U_S = U_T$.

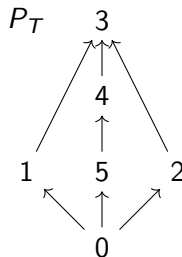
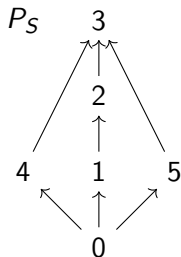
Thus U_S encodes the combinatorics of $\text{Ap}(S, m)$.

Corollary

The graph U_S determines the ideal $\frac{(X_0)+I_S}{(X_0)}$ up to a permutation of the variables.

Example (part 2)

Take S and T as in the previous example. Since



S and T lie in different faces of the Kunz cone, but, since $U_S = U_T$, S and T are similar.

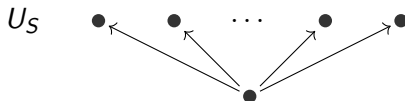
The graph of M.E.D. numerical semigroups

A numerical semigroup $S \in \mathcal{S}_m$ has **maximal embedding dimension** if S has exactly m minimal generators, or, equivalently, if the set

$$\text{Ap}(S, m) \setminus \{0\} \cup \{m\}$$

is the minimal set of generators for S .

In particular, for all $u, v \in \text{Ap}(S, m) \setminus \{0\}$ we have that $u \not\preceq_S v$ and thus



It follows that numerical semigroups with maximal embedding dimension form an equivalence class with respect to the relation of being similar.

Determinantal ideals are not licci

Ideals of R that are generated by maximal minors of a $2 \times n$ matrix are known to be non-licci.

Recall that $f \in \mathbb{N} \setminus S$ is a **pseudo-frobenius number** of S if

$$f + S \setminus \{0\} \subseteq S.$$

Theorem (Kien, Matsuoka, 2020)

If S a numerical semigroup with maximal embedding dimension, the following conditions are equivalent:

- *the pseudo-frobenius numbers of S form an arithmetic sequence of length $n - 1$;*
- *I_S is generated by the maximal minors of a $2 \times n$ matrix.*

Let $m \geq 4$. The numerical semigroup

$$S = \langle m, 2m+1, \dots, 3m-1 \rangle = \{0, m, 2m, \dots\}$$

has m minimal generators, and therefore S has maximal embedding dimension. Moreover, the pseudo-frobenius numbers of S are

$$\{m+1, \dots, 2m-1\}$$

and thus they form an arithmetic sequence. This implies that I_S is generated by the maximal minors of a $2 \times m$ matrix and thus I_S is not licci.

Numerical semigroups generated by 4 elements

Guerrieri, Ni and Weyman characterized licci ideals of height equal to 3 with methods coming from representation theory. They identify formats of free resolutions coming from the exceptional Lie algebras of type A, D and E. These formats are called **Dynkin formats** and they are

$$(1, 3, 3, 1) , (1, 5, 5, 1) , (1, 4, n, n - 3) \text{ for } n \geq 5, \\ (1, 5, 6, 2) , (1, 5, 7, 3) , (1, 6, 7, 2) , (1, 5, 8, 4) , (1, 7, 8, 2).$$

If the format of the minimal free resolution of R/I is Dynkin, then I is licci.

Experimental evidence suggests that the following holds:

Conjecture

Let S be a numerical semigroup minimally generated by 4 elements. The ideal I_S is licci if and only if the format of the minimal free resolution of R/I_S is a Dynkin format.

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- ② L. Guerrieri, X. Ni, J. Weyman, *Higher structure maps for free resolutions of length 3 and linkage*, Journal of Pure and Applied Algebra, vol. 230, no. 6 (2026)
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Thank you!